FIT Hon Teng Limited

鴻騰六零八八精密科技股份有限公司

CPD-X Test Station Introduction

2026/1/12

E&F SW Authentication

SW Planning

1. Verify & Plan ERS/DFM

2. Simulate test Item & communication

3. Extend Design Features

4. Let vendor preparing self-test program (Case QT)

5. Integrate test item of overlay ,e.g. Atlas /Python/Swift

Off-line Mode

1.Certificate Holder & Fixture condition with DFM

2.Run overlay and remove CB/UOP functions

3. SW team needs finishing 50up loop or GRR test

Case QT/WLC

Audit Mode

1.Test Vendor’s Holder & Fixture & overlay before P1 build

2.Put overlay on GHDS, remove CB/UOP

3. Need uploading test result to Insight

4. Don’t write the calibration value on acoustic station

Bud QT

Normal Mode

1.Receive test data between P1 to EVT build ,calculate the Retest Rate & Fair Rate on TIL report

2.Add CB/UOP items

3.Need uploading test result to Insight

4.Modify overlay to decrease retest rate & test time

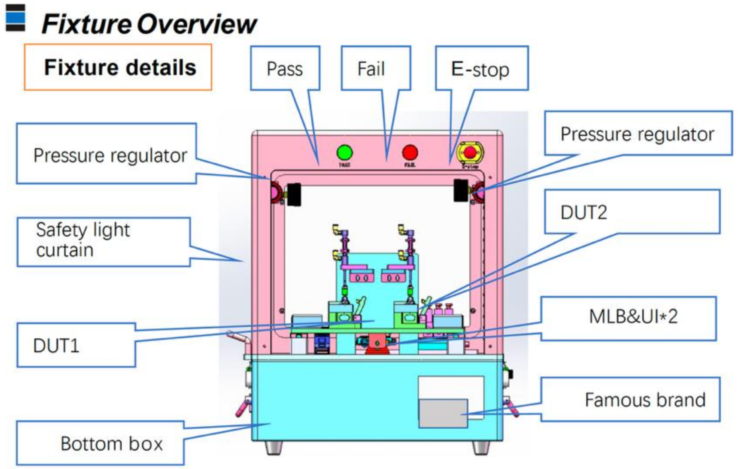
Bud QT Station Introduction

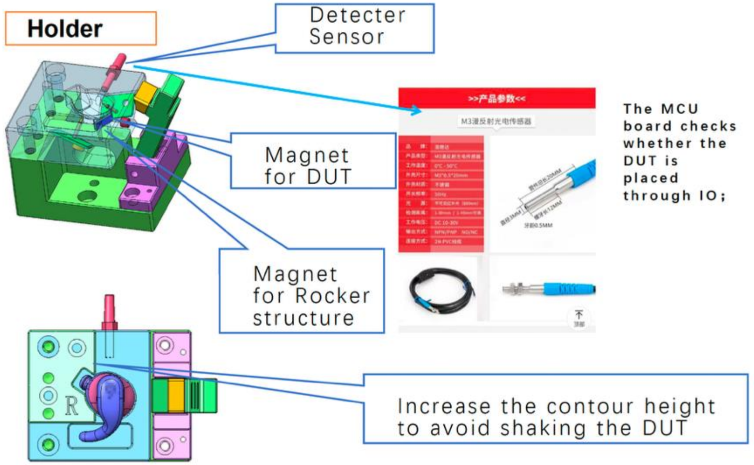
Test Model:B494b Vendor : 星禾宏泰/泰諾嘉

Objective :
- Write and upload DUT information (FATP SN, FATP CFG, battery information)
- Check Locust Connectivity
- Check BUCK/LDO Function
- Check Kessel & IMU Connectivity
- Check battery status
- Check button status

Brief introduction of main equipment :
- ✓ B494b system test fixture
- ✓ Control board module
- ✓ Cylinder module for DUT in /out and 1~3 button test
- ✓ Dut holder

Testing principle :
- Mac Mini communicates with DUT through grass ninja / B495 MLB, read and write DUT's information.
- Check the status of the Buck/LDO by reading the parameters of the corresponding test points.
- Read the chip information by sending a command to check the connectivity of the chip.
- Send commands to check DUT battery charger status.
- Use the cylinder to create the state of pressing and releasing to verify the function of the button.



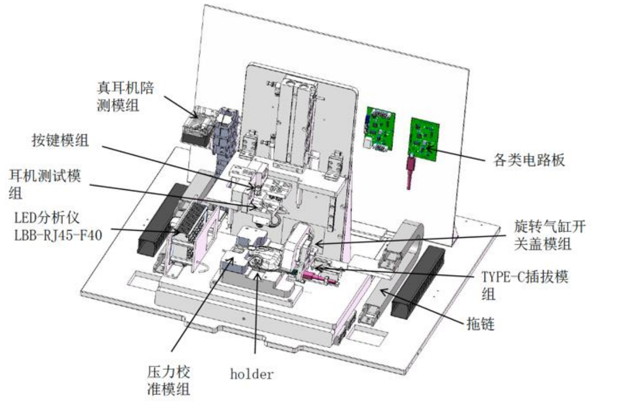


Terminology

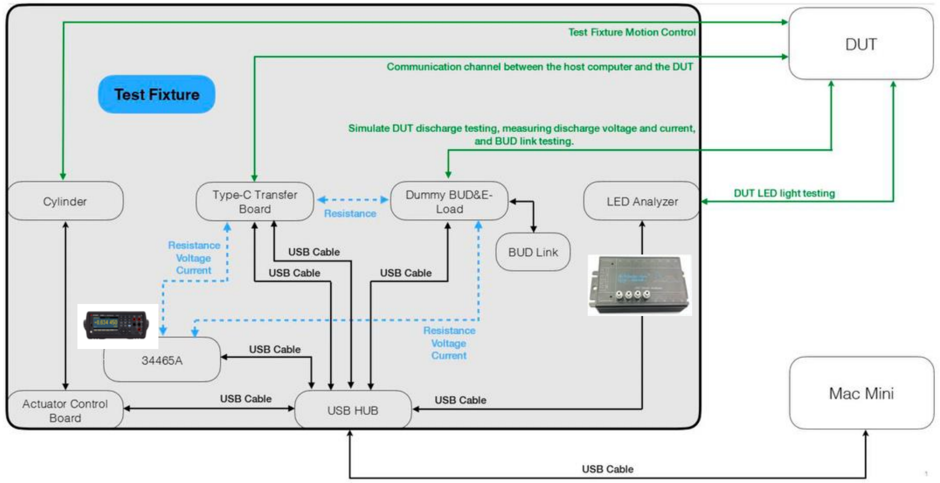
Test Items	Description
Bud Charging On	設定Charging On
Get Case Version	讀取Case FW/HW version
Button Test	降下、上拉圓柱棒來做按鈕按壓測試
Read Bud SN	讀取Bud MLB/FATP SN
Chack Battery Status	檢查電池資訊
Read BUCK and LDO	讀取 BUCK and LDO 1/2/3
Quiescent Current	自用電流
Check Status and Syscfg	檢查Kessel Type/RevID/Bud FW Version, Syscfg ACRT/SCRT
Check HWID/ECID	檢查HWID/ECID
Read VBATT	讀取電池電壓
IMU Test	慣性量測單元/加速度計/陀螺儀
Check OTP Locked	OTP鎖定
Enable UVP	啟動UVP
Check Bud Link	檢查Bud Link (此時以先Close Tunnel)

Case QT Station Introduction

Test Model:B499 Vendor : 東方恆越/上海盤雲



Objective: (test coverage)
•Test Case GND DCR.
•Write and Read Case information (FATP SN, FATP CFG, FW version, HWID, BTID)
•Check Case battery charger status and FuelGauge calibration information.
•Check Case signal indicate LED and button status.
•Test Case ship mode Enter & Exit.
•Test the voltage and current of Case output 5V under no-load/loaded conditions.
•Test the connectivity between Case and Bud.
•Check Case lid status.

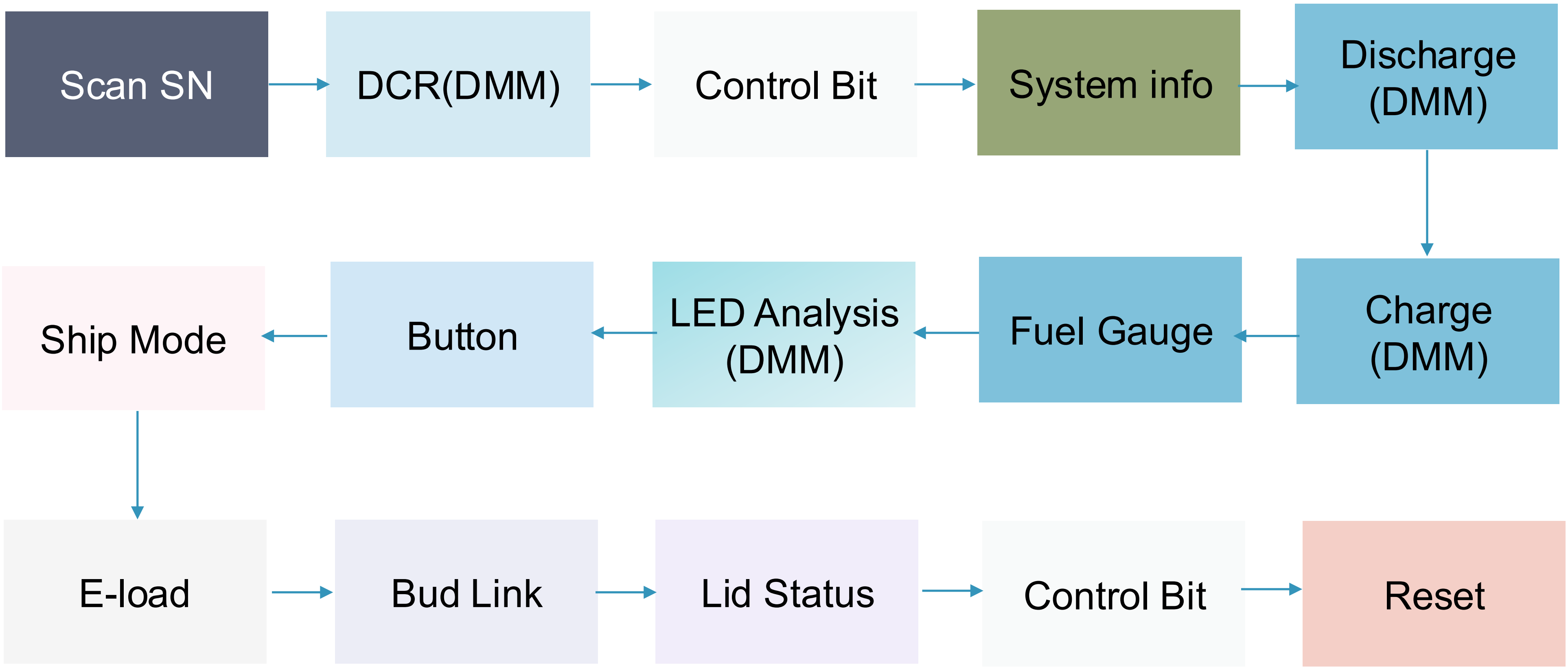


Brief introduction of main equipment
•Teye-C auto insertion module.
•Pusher /Puller module(東方恆越).
•Control Box module.
•Sliding in and out module.
•Holder with DUT module.
•LED Analyzer.
•DMM.

Testing principle
•Use DMM through dummy bud to measure GND DCR from Type-C to BUD_L/R_NEG.
•Mac Mini communicates with Case through Type-C cable, send commands to read and write Case information (FATP SN, FATP CFG, FW version, HWID, BTID).
•Send commands check Case battery charger status and FuelGauge calibration information.
•Use an LED analyzer to measure the X/Y color coordinates, brightness, and wavelength of the signal lights in the Case when they emit red and white light. Use the cylinder to create the state of pressing and releasing to verify the function of the button.
•Send command to put the Case into ship mode, disconnect and reconnect VBUS voltage to exit ship mode.
•Send command to the L/R POS point output 5V of the Case, and measure the voltage and current when loaded with 100mA and 0mA.
•Connect to B498 bud t and send a command to read TX and RX values to check the communication function between Case and bud.
•Use the cylinder to create the state of opening and closing to verify the function of the Case lid.

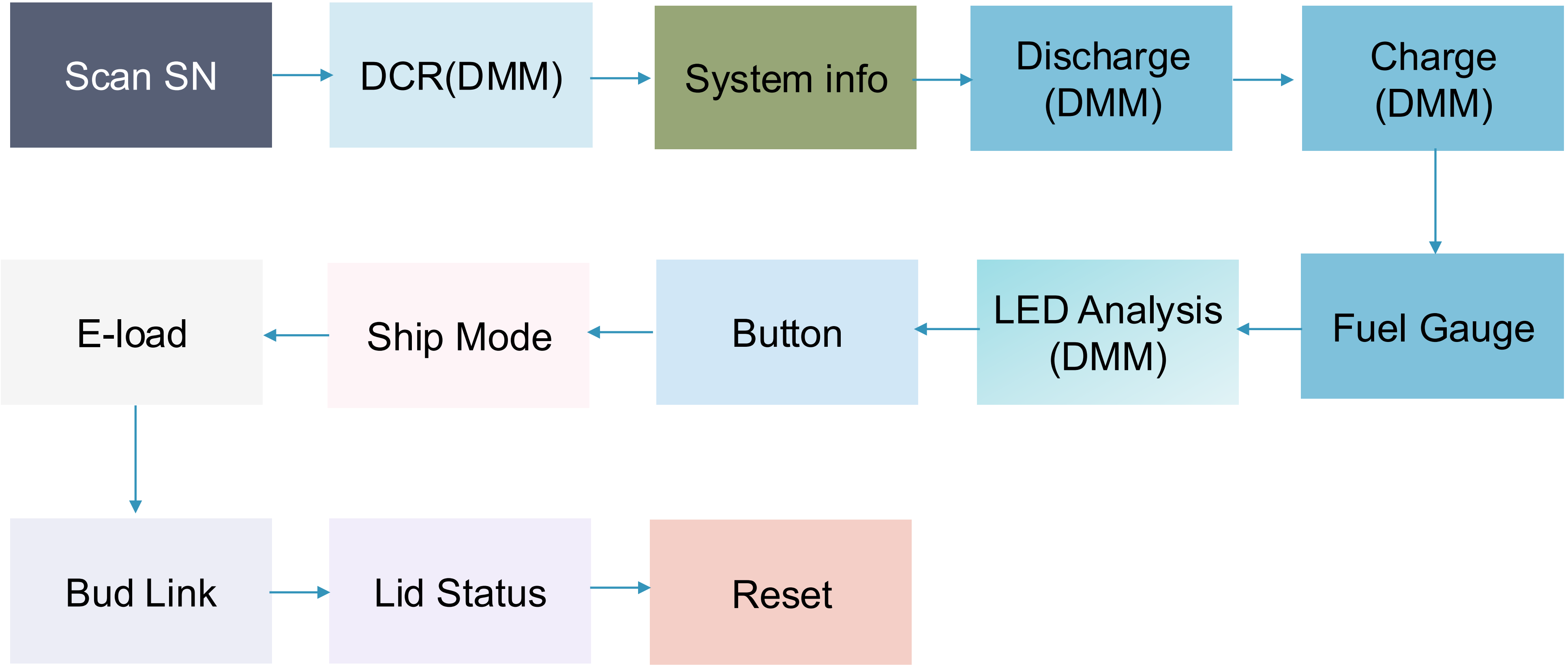
Terminology

Case QT - SW test flowchart(Online)：



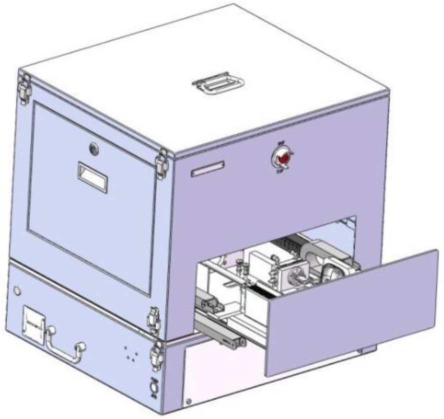
Terminology

Case QT - SW test flowchart：

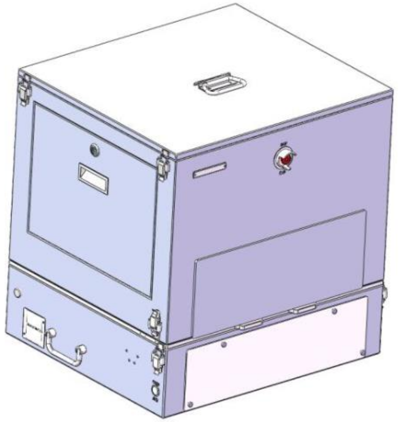


Terminology

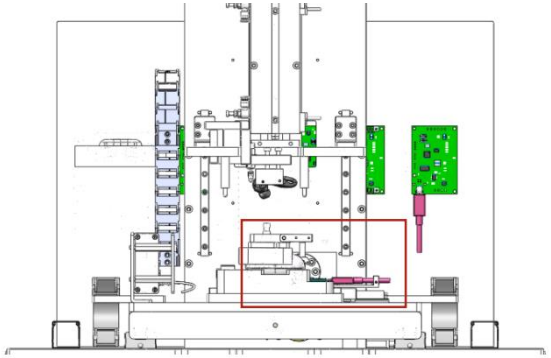
Case QT - test flowchart：



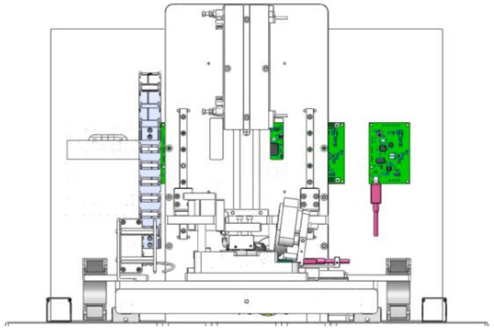
OP放入产品，按
下双手启动按钮



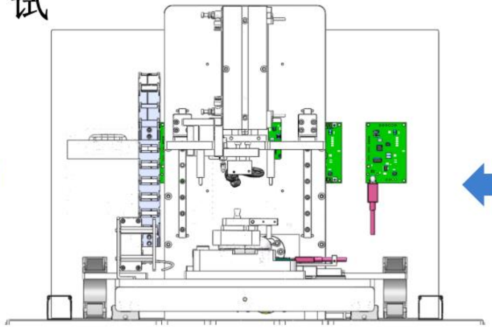
治具进仓，开始测
试



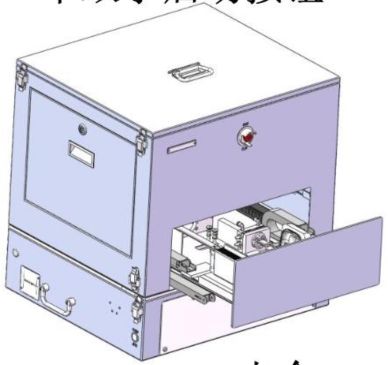
旋转气缸压合后插入TYPE-
C，然后旋转模组旋转开盖



上压模组下压，开始耳机和
按键测试



测试完毕旋转气缸关盖，拔
出TYPE-C



出仓

Terminology

Test Items	Description
scan_sn	Read and verify the SN
CloseDoor	Close the fixture door
DeviceReset	Reset the dummy board and Type-C board
TypeC_In	Insert the Type-C module
Press_DUT	Open the lid and press down to secure the DUT
DCR	Measure GND DCR from Type-C to BUD_L/R_NEG
DUTPowerON	Open DUT communication channel
ResetDUT	Find DUT USB port and reset
CleanDUTPort	Use UART to send a command and verify the DUT port communication
Bootmode	Set the Bootmode 2 to facilitate subsequent testing
MLBSN	Read and verify the MLBSN
SNWrite_SrNm	Read the SN and Write FGSN
SrNm	Read the SN and FGSN and verify that they match

Terminology

Test Items	Description
BoardID	Read and verify the HWID and PID
FW	Read and verify the FW version
BTID	Read and verify the BTID
USB	Get the USB information (Product / Vendor ID)
PowerMux	Read GPIO Data
BatteryDisChargeStatus	Verify voltage, current, SOC, and temp when the Case charger disable
BatteryChargeStatus	Verify voltage, current, SOC, and temp when the Case charger enable
Info	Check battery and FuelGauge calibration information
Brightness	Measure the Case LED parameters under red and white light conditions
GPIOCheck_Button_(Status)	Verify the ststus of the button under the state of pressing and releasing
OVP	Verify that the VBUS voltage is normal when OVP is disabled
Reset	Verify the DUT reset behavior upon VBUS removal
Transition	Test Case ship mode Enter & Exit
Constant	Measure the output voltage & current at the L/R POS under load conditions

Terminology

Test Items	Description
DummyReset	Reset the Dummy board
ConnectivityTest_PARROT	Verify the connectivity between the DUT and the PARROT device
OpenBudlinkChannel	Open the dummy board budlink channel
ConnectivityTest_BuddyLink(Right/Left)	Check the communication function between Case and bud
CloseBudlinkChannel	Close the dummy board budlink channel
Release_DUT	Raise the dummy-bud pressing module to release the DUT fixture
LidStatusCheck	Check Case lid status
SystemReset	Reset the DUT device
Teardown	
FixtureReset	Reset the fixture to ensure it returns to the initial position
cleanSN	Clear the DUT's SN to allow reprogramming or retesting.

Terminology

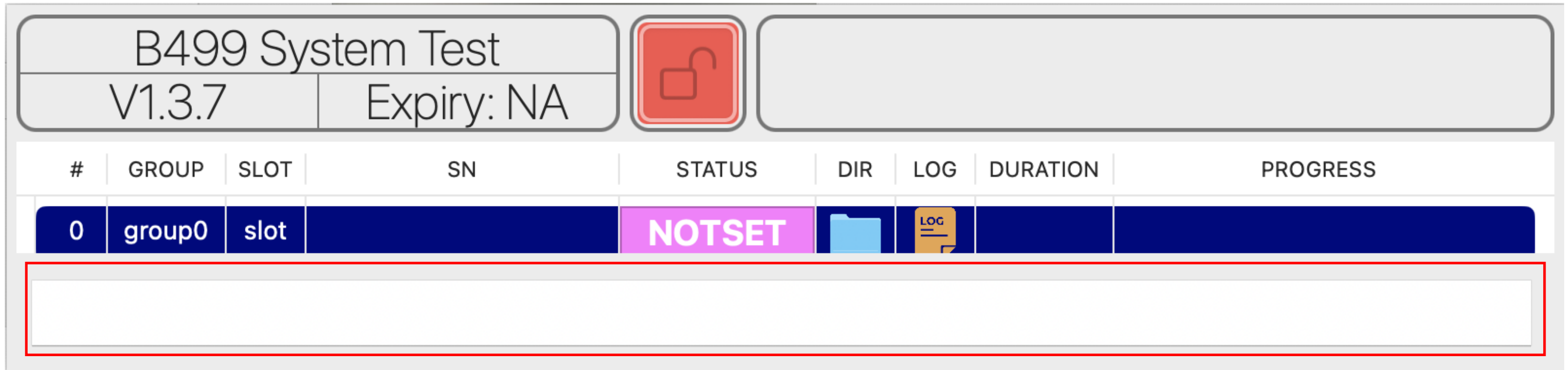
Cylinder Board Command :

Functions	Command	Return	Notes
開始測試		START	按下左右按鈕，且氣壓缸有正確吸取時才發送 START
進出汽缸工作位(入)	cylinder1 front	cylinder1 front[ok]	先關蓋汽缸動作(cylinder3 front)，再進入汽缸工作位
進出汽缸初始位(外)	cylinder1 back	cylinder1 back[ok]	初始位後，底部真空吸關(cylinder6 back)
Type-C汽缸動作	cylinder2 front	cylinder2 front[ok]	
Type-C汽缸復位	cylinder2 back	cylinder2 back[ok]	
關蓋汽缸動作	cylinder3 front	cylinder3 front[ok]	
關蓋汽缸復位	cylinder3 back	cylinder3 back[ok]	開蓋真空吸開後復位
開關蓋狀態判定	Check DUT	DUT[n]	DUT[0]代表關蓋、DUT[1]代表開蓋
假耳下壓汽缸動作	cylinder4 front	cylinder4 front[ok]	
假耳汽缸復位	cylinder4 back	cylinder4 back[ok]	
按鍵汽缸動作	cylinder5 front	cylinder5 front[ok]	
按鍵汽缸復位	cylinder5 back	cylinder5 back[ok]	
底部真空吸開	cylinder6 front	cylinder6 front[ok]	綁在cylinder1 front前
底部真空吸關	cylinder6 back	cylinder6 back[ok]	綁在cylinder1 back後
開蓋真空吸開	cylinder9 front	cylinder9 front[ok]	新設計
開蓋真空吸關	cylinder9 back	cylinder9 back[ok]	新設計
測試成功	Pass	Pass OK	測試成功燈號
測試失敗	Fail	Fail OK	測試失敗燈號
reset	reset	Reset OK\r\n	全部動作復位
help	help		Command list

Terminology

scan_sn：

- Read and verify the SN



Use TPInterview Plugin to get the SN

```lua
-- if #sn ~= 12 then
if #sn == 12 or #sn == 10 then
    bResult = true
else
    bResult = false
end
```

SN length must be 12 or 10

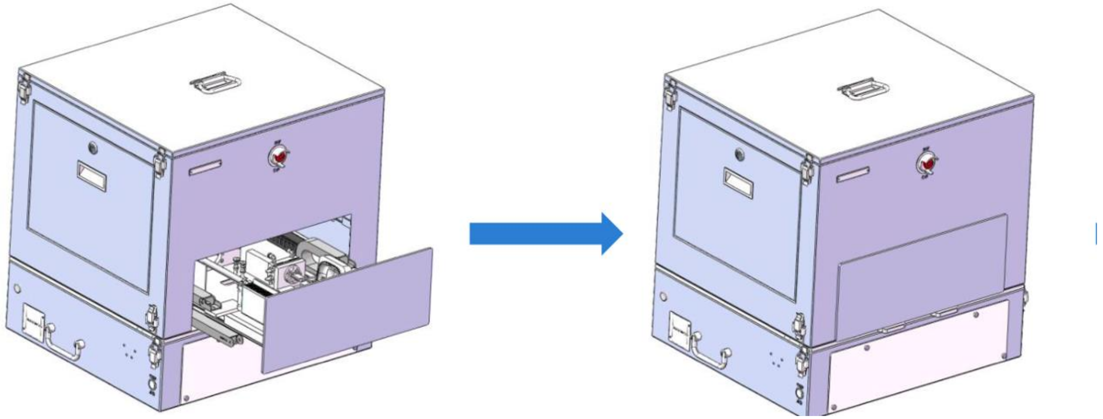
Terminology

CloseDoor：

- Close the fixture door

Command：

Test item	cylinder command	Description
CloseDoor	cylinder1 front	進入工作位（先關蓋）



Terminology

DeviceReset：

- Reset the dummy board and Type-C board

Command：

Test item		Type-C command	Description	Dummy command	Description
DeviceReset	1	reset	復位	reset	復位

Terminology

TypeC_In：

- Insert the Type-C module

Command：

Test item	cylinder command	Description
TypeC_In	cylinder2 front	進入工作位（先關蓋）

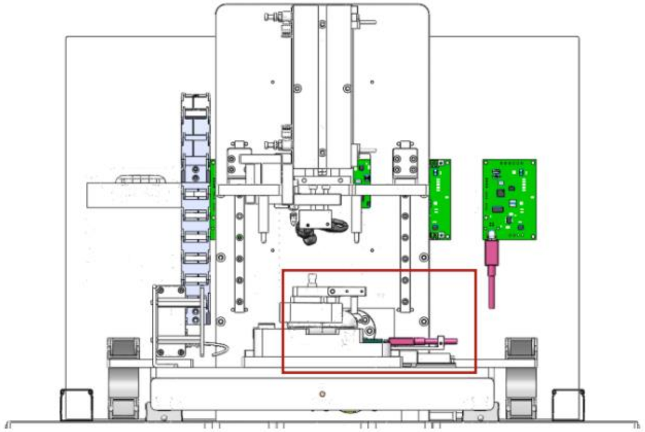
Terminology

Press_DUT：

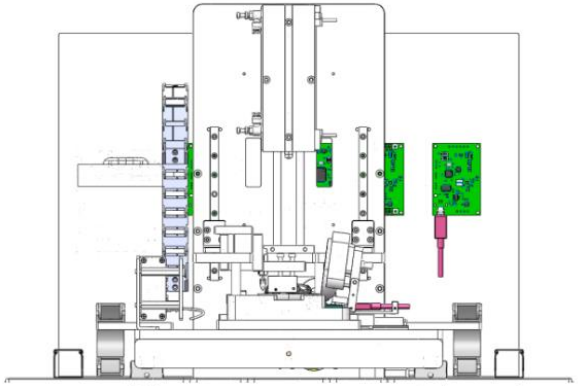
- Open the lid and press down to secure the DUT

Command：

Test item		cylinder command	Description
Press_DUT	1	cylinder3 back	關蓋汽缸復位(開蓋)
	2	Check DUT	開關蓋狀態判定
	3	cylinder4 front	假耳下壓動作



旋转气缸压合后插入TYPE-C，然后旋转模组旋转开盖



上压模组下压，开始耳机和按键测试

Terminology

DCR：

- Measure GND DCR from Type-C to BUD_L/R_NEG

Command：

Test item		Type-C cmd	Description	Dummy cmd	Description	shell cmd	Description	Cylinder cmd	Description
DCR (Left /Right)	1	test_vbus_contact_resistance_on	打開通道,DMM獲取內阻						
	2			test_(L/R)_contact_resistance_on	打開通道,DMM獲取內阻				
	3					DMM.py	量測DCR(1st)		
	4							cylinder4 back	假耳上抬
	5							cylinder4 front	假耳下壓
	6					DMM.py	量測DCR(2nd)		
	7			test_(L/R)_contact_resistance_off	測試完成,關閉通道				
	8	test_vbus_contact_resistance_off	測試完成,關閉通道						

Terminology

DUTPowerON：

- Open DUT communication channel

Command：

Test item		Type-C command	Description	Dummy command	Description
DUTPowerON	1	PC_MODE_ON	打開DUT通信通道（USB）		
	2	vbus[1][5]	設定外接電源5v		

Terminology

ResetDUT：

- Find DUT USB port and reset

Command：

Test item		Shell command	Description	DUT_USB command	Description	DUT command	Description
ResetDUT	1	find /dev/cu.usbmodem*	macOS 找 DUT serial port				
	2			\n			
	3			ft reset\n	重置DUT		
	4					ft fw ver	當回應是亂碼才執行

若連接失敗，執行以下指令

Test item		cylinder command	Description
ResetDUT	1	cylinder2 back	Type-C汽缸復位
	2	cylinder2 front	Type-C汽缸動作

Retry 最多 3次

Terminology

CleanDUTPort：

- Use UART to send a command and verify the DUT port communication

Command：

Test item		DUT command	Description
CleanDUTPort	1	ft fw ver	透過回傳，確保連線是正常的

Terminology

Bootmode：

- Set the Bootmode 2 to facilitate subsequent testing

Command：

Test item		DUT command	Description
Bootmode	1	ft bootmode set 2	設定bootmode為2(SMT 工廠模式)
	2	ft bootmode get	獲取bootmode,檢查設定是否成功

```
Case) .ft bootmode get
0x2
> ft:ok
Case]
```

Get Bootmode 2

Terminology

MLBSN：

- Read and verify the MLBSN

Command：

Test item		DUT command	Description
MLBSN	1	ft sn read mlb	讀取mlbsn

```lua
if string.find(stringResp,"ft:ok") then
    local data = utils.cutString(stringResp,"ft:ok ","\n")
    mlbsn = data
    if #mlbsn > 10 then
        bResult = true
    end
end
```

確認回傳字串並驗證mlbsn

Terminology

SNWrite_SrNm：

- Read the SN and Write FGSN

Command：

Test item		DUT command	Description
SNWrite_SrNm	1	ft sn read system	讀取盒子sn 確認是否與掃描的sn相同
	2	ft sn write system	若盒子無sn 則寫入掃描時的sn

在傳送指令前會先確認sn有值才開始

Terminology

SrNm：

- Read the SN and FGSN and verify that they match

Command：

Test item		DUT command	Description
SrNm	1	ft sn read system	讀取FGSN

Terminology

BoardID：

- Read and verify the HWID and PID

Command：

Test item		DUT command	Description
BoardID	1	ft hw ver	讀取硬體版本
	2	adc read pid	讀取硬體識別電壓

Terminology

FW：

- Read and verify the FW version

Command：

Test item		DUT command	Description
FW	1	ft fw ver	讀取韌體版本

```
[Main-15-1][][_DUTDevice.lua][INFO]  retStr is :Case] ft fw ver
> ft:ok begin
  Active   APPS Version: 2.9.2 ((B&I) 2025-03-14 01:26:01 PDT)
  Inactive APPS Version: 2.9.2 ((B&I) 2025-03-14 01:26:01 PDT)
  Active Bank:       1
  Inactive Bank:     2
> ft:ok end
Case]
```

讀取驗證FW active 和 inactive version

Terminology

BTID：

- Read and verify the BTID

Command：

Test item		DUT command	Description
BTID	1	ft btid read	讀取bitd
	2	ft btid write 1	若沒讀取到btid才寫入
	3	ft btid read	驗證bitd

Terminology

USB：

- Get the USB information (Product / Vendor ID)

Command：

Test item		shell command	Description
USB	1	system_profiler SPUSBDataType	取得USB 裝置的詳細資訊

```
Beats Fit Pro Case:

  Product ID: 0x4022
  Vendor ID: 0x290b
  Version: 1.06
  Serial Number: GMROffDUT3
  Speed: Up to 12 Mb/s
  Manufacturer: Beats Electronics
  Location ID: 0x14200000 / 14
  Current Available (mA): 500
  Current Required (mA): 500
  Extra Operating Current (mA): 0
```

讀取驗證Product / Vendor ID

Terminology

PowerMux：

- Read GPIO data

Command：

Test item		DUT command	Description
PowerMux	1	gpio get A 9	讀取GPIO 晶片資料

Terminology

Battery(DisCharge / Charge)Status：

- Verify voltage, current, SOC, and temp when the case charger (disable / enable)

Command：

Test item		DUT command	Description	Type-C command	Description
BatteryDisChargeStatus	1	charger disable	關閉充電功能		
BatteryChargeStatus	1	charger enable	開啟充電功能		
Battery(DisCharge/Charge)Status	2	charger regdump	讀取充電晶片暫存器值		
	3	gg info	確認電池各項資訊		
	4			getvoltage	獲取電壓
	5			getcurrent	獲取電流
	6	charger vbus status	確認vbus狀態		
	7	charger adc read vbus	讀取 VBUS 電壓		
	8	charger adc read ibus	讀取 VBUS 電流		
	9	charger adc read vbat	讀取 充電盒電池端電壓		
	10	charger adc read ibat	讀取 充電盒電池充放電電流		
	11	ft temperature get	檢查MCU溫度		
	12	charger adc read TDIE	檢查電池溫度		

Terminology

Info：

- Check battery and fuel gauge calibration information

Command：

Test item		DUT command	Description
Info	1	gg unseal	解除fuel gauge校正值鎖定
	2	gg cal info	取得fuel gauge校正值資訊
	3	gg info	確認電池各項資訊
	4	gg cc_offset_cal	設定fuel gauge校正值
	5	gg seal	鎖定fuel gauge校正值
	6	gg reset	reset

```
> gg:ok begin
  Smooth SoC: 64 %
  Raw SoC: 66 %
  Voltage: 4044 mV
  Temperature: 25.7 C
  Avg Current: 0 mA
> gg:ok end
```

gg info

Terminology

Brightness：

- Measure the case LED parameters under red and white light conditions

Command：

Test item		DUT command	Description	LED command	Description	Type-C command	Description
	1	dev vbus enable	開啟vbus供電				
	2	charger disable	關閉充電功能				
	3	pwm unset	關閉盒子燈				
	4					getcurrent	獲取LED無光vbus電流
	5			:001w_target_type01-01=0	設定LED分析儀測白光		
	6			:001w_offset_en01-01=1	設定白光校正補償		
	7			:001r_chroma01-01	讀取數值（無光）		
	8	PWM set white 1000	設定盒子白光				
Brightness	9					getcurrent	獲取LED白光vbus電流
	10			:001r_chroma01-01	讀取數值（白光）		
	11	pwm unset	關閉盒子燈				
	12	PWM set red 1000	設定盒子紅光				
	13			:001w_target_type01-01=11	設定LED分析儀測紅光		
	14			:001w_offset_en01-01=2	設定紅光校正補償		
	15					getcurrent	獲取LED紅光vbus電流
	16			:001r_chroma01-01	讀取數值（紅光）		
	17	pwm unset	關閉盒子燈				

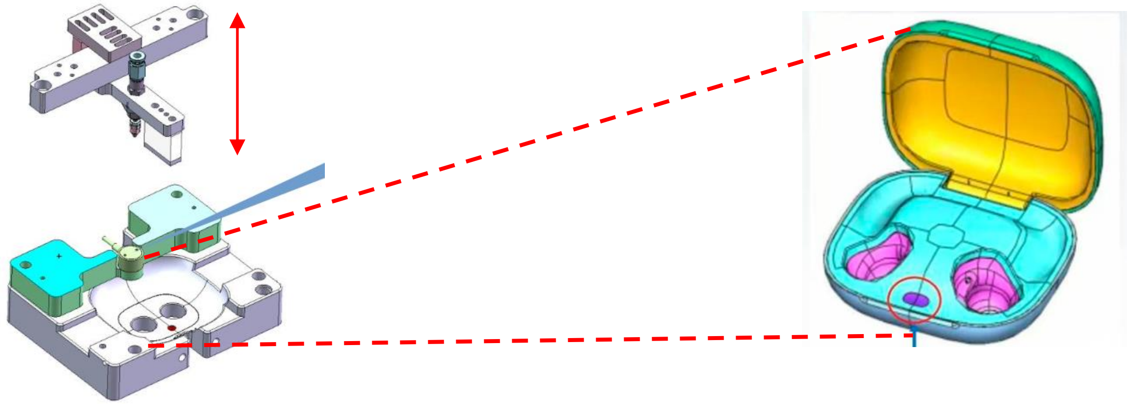
Terminology

GPIOCheck_Button_(Initial / Pressed / Released)：

- Verify the status of the button under the state of pressing and releasing

Command：

Test item		DUT command	Description	cylinder command	Description
GPIOCheck_Button_Initial	1	dev btn get	讀取按鍵狀態（釋放）		
GPIOCheck_Button_Pressed	1			cylinder5 front	按鍵汽缸往前按壓
	2	dev btn get	讀取按鍵狀態（按壓）		
GPIOCheck_Button_Released	1			cylinder5 back	按鍵汽缸往後鬆開
	2	dev btn get	讀取按鍵狀態（釋放）		



Terminology

OVP (Over Voltage Protection)：

- Verify that the VBUS voltage is normal when OVP is disabled

Command：

Test item		DUT command	Description
OVP	1	dev vbus disable	關閉過壓保護
	2	dev vbus get	檢查vbus電壓

Terminology

Transition：

- Test case ship mode Enter & Exit

Command：

Test item		DUT command	Description	Type-C command	Description
Transition	1			vbus[1][5]	設定外接電源5v
	2	dev powermode set standby	設定充電盒休眠模式		
	3			vbus[1][0]	設定外接電源0v (斷連)
	4			vbus[1][5]	設定外接電源5v (重連，離開休眠)

Terminology

Constant：

- Measure the output voltage & current at the bud L/R POS under (no/ load) conditions

Command：

Test item		DUT command	Description	Dummy command	Description
Constant	1	charger disable	關閉充電功能		
	2	dev vbus disable	停用5v外部電源		
	3	dev vbus get	確認停用vbus		
	4	charger boost enable	啟用 Boost Mode升壓供電		
	5	charger boostv set 5000	設定 Boost 輸出電壓5 V		
	6	bud mgr disable	停用 bud的管理		
	7	bud chg set left on	打開 左耳充電路徑		
	8	bud chg set right on	打開 右耳充電路徑		
	9			set_left_eload[0]	設定 左耳電子負載電流 0mA
	10			set_right_eload[0]	設定 右耳電子負載電流 0mA
	11			left_eload_ch_on	打開左耳Eload通道
	12			right_eload_ch_on	打開右耳Eload通道
	13			set_left_eload[100]	設定 左耳電子負載電流 100mA

Terminology

Constant：

- Measure the output voltage & current at the bud L/R POS under load conditions

Command：

Test item		DUT command	Description	Dummy command	Description
Constant	14			set_right_eload[100]	設定 右耳電子負載電流 100mA
	15			get_left_pos_voltage	讀取 左耳 Bud Pogo pin 電壓
	16			get_left_eload_current	讀取 左耳電子負載實際的電流
	17			get_right_pos_voltage	讀取 右耳 Bud Pogo pin 電壓
	18			get_right_eload_current	讀取 右耳電子負載實際的電流
	19	gg info	取得電池各項資訊		
	20			set_left_eload[0]	設定 左耳電子負載電流 0mA
	21			set_right_eload[0]	設定 右耳電子負載電流 0mA
	22			get_left_pos_voltage	讀取 左耳 Bud Pogo pin 電壓
	23			get_right_pos_voltage	讀取 右耳 Bud Pogo pin 電壓
	24	gg info	確認電池各項資訊		
	25			（Left / Right）_eload_ch_off	關閉左右耳Eload通道
	26	dev vbus enable	重新啟用5v外部電源		
	27	dev vbus get	確認啟用vbus		

Terminology

ConnectivityTest_PARROT：

- Verify the connectivity between the DUT and the PARROT device

Command：

Test item		DUT command	Description
ConnectivityTest_PARROT	1	I2c scan i2c1	在 i2c1 bus 上掃描所有裝置位址

Terminology

OpenBudlinkChannel & CloseBudlinkChannel：

- Reset the fixture to ensure it returns to the initial position
- Clear the DUT's SN to allow reprogramming or retesting.

Command：

Test item		Dummy command	Description
OpenBudlinkChannel	1	right_budlink_ch_on	打開右耳連接通道
	2	left_budlink_ch_on	打開左耳連接通道
CloseBudlinkChannel	1	right_budlink_ch_off	關閉右耳連接通道
	2	left_budlink_ch_off	關閉左耳連接通道

Terminology

ConnectivityTest_BuddyLink(Right/Left)：

- Check the communication function between case and bud

Command：

Test item		DUT command	Description
ConnectivityTest_BuddyLink (Right/Left)	1	ft reset	reset系統
	2	dbg set all none	Debug 相關設定設為無
	3	bud mgr disable (left/right)	停用（L/R）Bud的管理
	4	bud detect (left/right)	偵測（L/R）Bud
	5	bud mgr enable (left/right)	啟用（L/R）Bud的管理
	6	bud state get (left/right)	獲取（L/R）Bud目前狀態
	7	bud link test (left/right)	測試（L/R）Bud連線功能

```
> bud:ok DETECTED
Case]
```

bud detect

```
  TX: 316, RX: 316 (100%)
  TXretry: 0, RXerr: 0, MPto:0, RSPto:0
> bud:ok
Case]
```

bud link test

Terminology

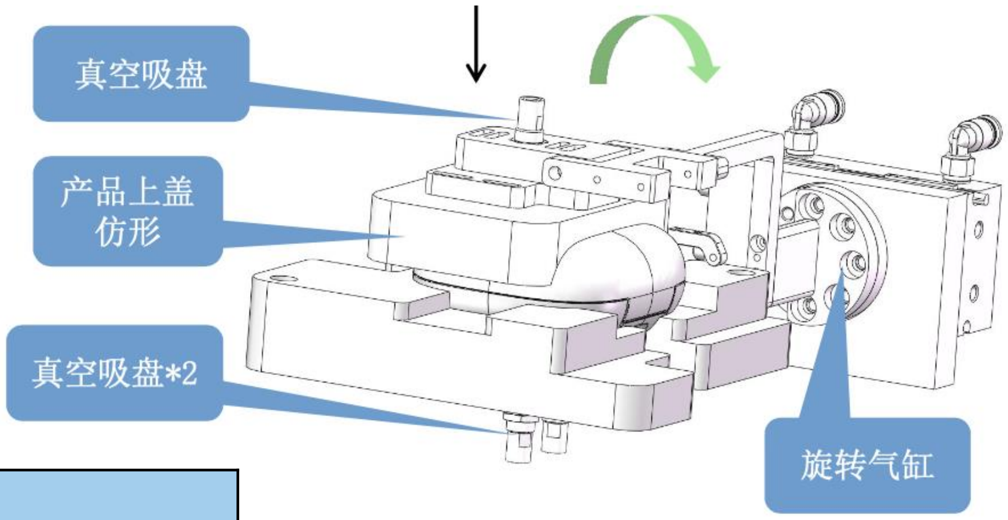
Release_DUT：

- Raise the dummy-bud pressing module to release the DUT fixture

Command：

Test item		cylinder command	Description
Release_DUT	1	cylinder4 back	假耳汽缸上抬

Terminology

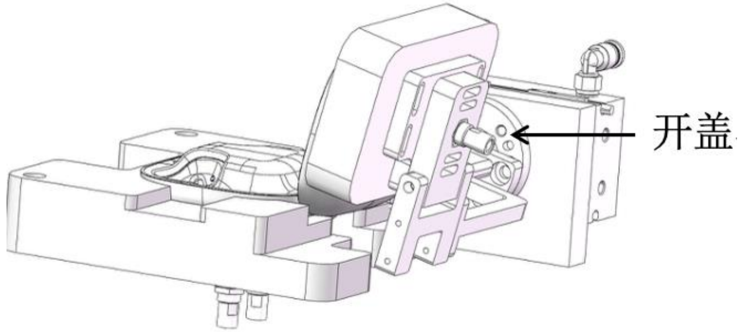


LidStatusCheck：

- Close the lid and Check case lid status

Command：

Test item		DUT command	Description	cylinder command	Description
LidStatusCheck	1	dev lid get	讀取盒子開關蓋狀態(開)		
	2			cylinder3 front	關蓋汽缸往前（關）
	3			cylinder3 back	關蓋汽缸往後（動作收回）
	4	dev lid get	讀取盒子開關蓋狀態(關)		



Terminology

SystemReset：

- Reset the DUT device

Command：

Test item		DUT command	Description
SystemReset	1	ft reset	重置待測裝置

Terminology

Teardown - FixtureReset ＆ cleanSN：

- Reset the fixture to ensure it returns to the initial position
- Clear the DUT's SN to allow reprogramming or retesting.

Command：

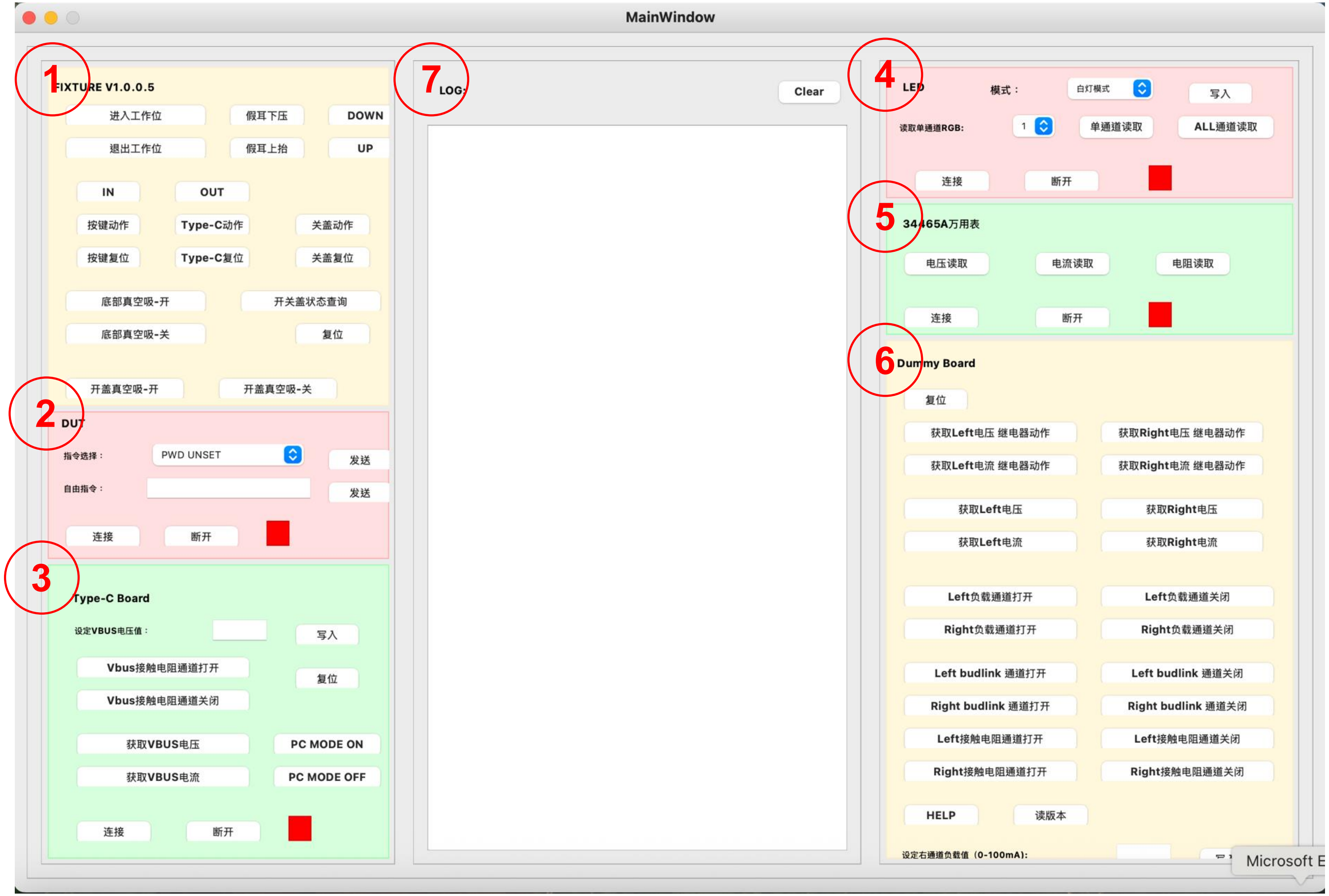
Test item		Type-C command	Description	Dummy command	Description	cylinder command	Description
FixtureReset	1	reset	復位	reset	復位		
	2					cylinder4 back	假耳汽缸上抬
	3					cylinder2 back	Type-C復位
	4					reset	全汽缸復位
cleanSN	1	TPinterview clear SN					

Terminology

Case QT - Debug Tool

介面顯示治具各模組連結狀態，並提供模組對應之功能，以供調適

1. Fixture
2. DUT
3. Type-C Board
4. LED Analysis
5. DMM-34465A
6. Dummy Board
7. Log display



Terminology

Case QT - Debug Tool - Fixture

Functions	Command
進入工作位（先關蓋）	cylinder1 front
退出工作位（後關下真空）	cylinder1 back
Type-C動作	cylinder2 front
Type-C復位	cylinder2 back
關蓋動作	cylinder3 front
關蓋復位	cylinder3 back
開關蓋狀態查詢	Check DUT
假耳下壓動作	cylinder4 front
假耳上抬動作	cylinder4 back
按鍵動作	cylinder5 front
按鍵復位	cylinder5 back
底部真空吸-開	cylinder6 front
底部真空吸-關	cylinder6 back
開蓋真空吸-開	cylinder9 front
開蓋真空吸-關	cylinder9 back
ＩＮ（僅進入工作位 不吸氣）	ＩＮ
ＯＵＴ（僅退出工作位）	ＯＵＴ
ＤＯＷＮ	同假耳下壓
ＵＰ	同假耳上抬
復位	reset

測試執行順序：

進入工作位 -> Type-C動作 ->關蓋復位(開蓋) -> 假耳下壓動作 -> 按鍵動作 -> 按鍵復位 ->假耳上抬動作 ->關蓋動作 ->假耳上抬動作(Teardown) -> Type-C復位(Teardown) ->復位

Terminology

Case QT - Debug Tool - DUT

Command	functions
pwm unset	關閉盒子燈
PWM set white 1000	設定盒子白光
PWM set red 1000	設定盒子紅光
Read MLB SN	非正式Case指令，對應指令參考 **Case QT设备继电器及气缸指令说明-V1.0 20251222.pdf**
Read FG SN	
Read BTID	
Read DUT HW ID	
Read DUT FW	
Read DUT PID	
GG INFO	
GG CAL INFO	
GG RESET	
READ BUTTON STATUS	
FT RESET	

預設

Terminology

Case QT - Debug Tool – Type-C Board

Functions	Command	Note
設定vbus電壓值	vbus[1][m]	m電壓值(0 / 5)V
Vbus接觸電阻通道打開	test_vbus_contact_resistance_on	
Vbus接觸電阻通道關閉	test_vbus_contact_resistance_off	
獲取vbus電壓	getvoltage	
獲取vbus電流	getcurrent	
PC MODE ON	PC_MODE_ON	開啟DUT USB通訊
PC MODE OFF	PC_MODE_OFF	調適用指令
復位	reset	

Terminology

Case QT - Debug Tool – LED Analysis

Functions	Command	Note
白燈模式	:001w_target_type01-01=0	設定LED分析儀測白光
	:001w_offset_en01-01=1	設定白光校正補償
紅燈模式	:001w_target_type01-01=11	設定LED分析儀測紅光
	:001w_offset_en01-01=2	設定紅光校正補償
獲取單通道RGB	:001r_chroma01-01	使用LED分析儀通道口，預設1
ALL通道獲取		全通道讀取

Terminology

Case QT - Debug Tool – DMM

Functions	Command	Note
電壓讀取	Use DMM.py	
電流讀取		
電阻讀取		測DCR

Terminology

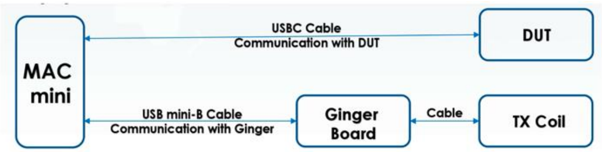
Case QT - Debug Tool – Dummy board

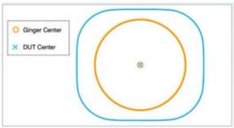
Functions	Command	Note
（L/R）電壓繼電器動作	get_(L/R)_pos_voltage_relay	調適用指令
（L/R）電流繼電器動作	get_(L/R)_eload_current_relay	調適用指令
獲取（L/R）電壓	get_(L/R)_pos_voltage	
獲取（L/R）電流	get_(L/R)_eload_current	
（L/R）負載通道打開	(L/R)_eload_ch_on	
（L/R）負載通道關閉	(L/R)_eload_ch_off	
（L/R）budlink通道打開	(L/R)_budlink_ch_on	
（L/R）budlink通道關閉	(L/R)_budlink_ch_off	
（L/R）接觸電阻通道打開	test_(L/R)_contact_resistance_on	
（L/R）接觸電阻通道關閉	test_(L/R)_contact_resistance_off	
設定（L/R）通道負載值	set_(L/R)_eload[n]	n負載值(0/100)mA

💡: (L/R)代表(left / right)

Wireless Charger Station Introduction

Test Model:B499 Vendor : 燦華

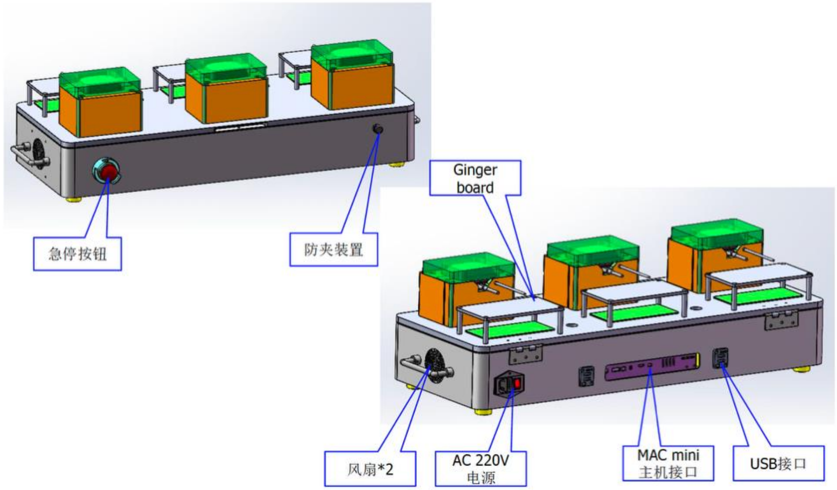




Objective: (test coverage)

•The purpose of this station is to test the wireless power transfer (WPT) of Bootmode 2 case

•Detect the test parameters of DUT at charging current of 40mA,120mA, 240mA, 360mA, 440mA and 480mA



Brief introduction of main equipment

✓ Wireless charging fixture (3up)
✓ Data Dependency tech , increase TT efficiency to 21.6%
✓ WASP(coil) and Ginger board : Composition TX simulator

Testing principle

•**Overview of Overlay Testing Principle:**

WASP (coil) and Ginger board open TX mode and set a fixed TX frequency.

Set wireless charging mode for DUT, set different charging currents(40ma~480ma), and read wireless charging test parameters for DUT.

•**Operation steps:**

OP connects the DUT to type-c, then places the DUT in the holder, and then places the pressure block on the holder to fix the DUT.

Overlay begins testing, and after testing is complete, remove the DUT from the holder

•**Main parameters of DUT wireless charging test:**

charger ADC value; gg info(voltage, current); wlchgr info(VRECT, ICUR) ; ginger power; Power Transfer Efficiency.

Terminology

Ginger_Set_TX	Set Ginger Board to TX Mode
Ginger_Set_coil_off	Disable TX coil output on the Ginger Board
Ginger_Get_dc_power	Measure DC input power when TX coil is disabled
Charger_enable	Enable the DUT charger on
Charger_Set_limits	Set and verify charger limit type to voltage mode
Charger_Set_Topoff_limits	Set and verify charger top-off voltage limits
PowerTransfer_BatCharging_40mA	Set case current, enable coil, verify wireless charger, read charger & battery info

Terminology

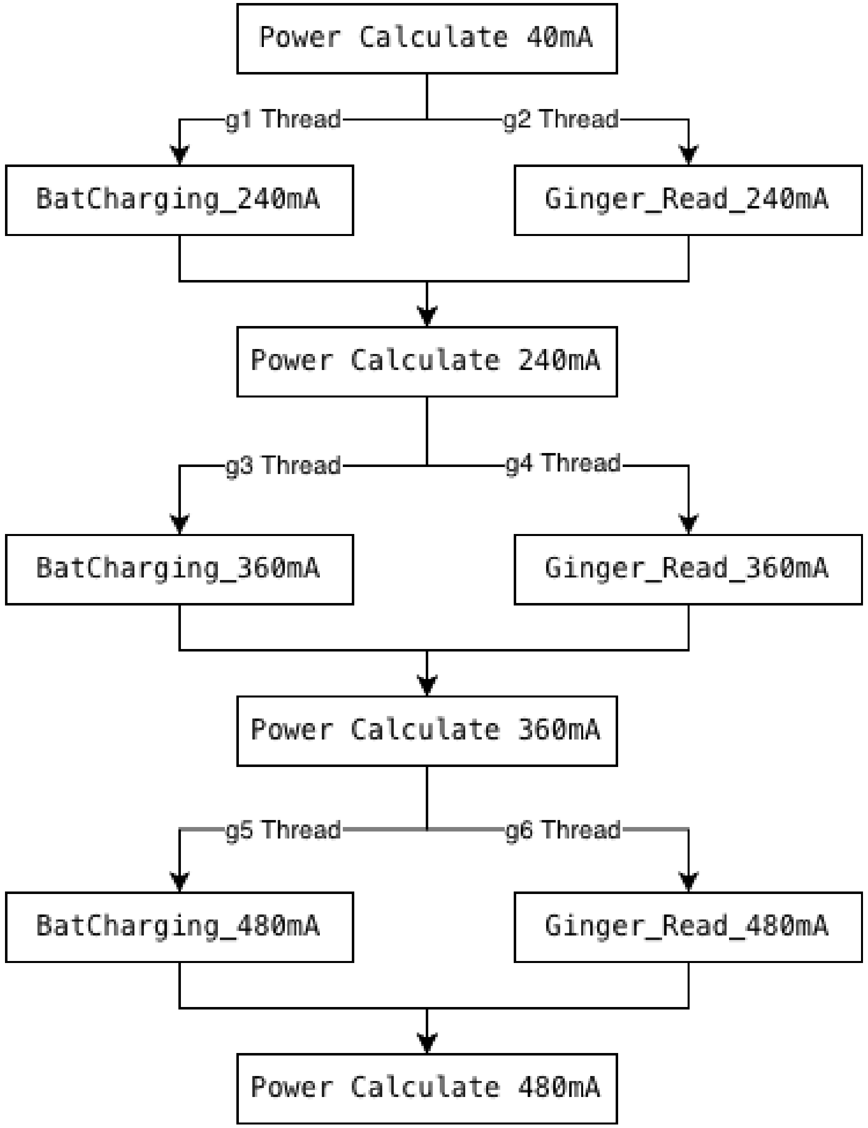
PowerTransfer_BatCharging_ 240 / 360 / 480 mA	Set case current and read wireless charger & battery info
Transition_Ginger_Read_240 / 360 / 480 mA	Read inverter DC input Power / Voltage/ Current
Transition_Ginger_Power_240 / 360 / 480 mA	Calculate Ginger board input power and verify transfer efficiency

Assets > Main.csv > data

```
TestName,Technology,Disable,Production,Audit,Thread,Policy,Loop,Sample,SOF,Condition,Notes
Transition_Ginger_Power_40mA,Process,,Y,Y,,,,,,,
PowerTransfer_BatCharging_240mA,Fwdl,,Y,Y,g1,,,,,,
Transition_Ginger_Read_240mA,Process,,Y,Y,g2,,,,,,
Transition_Ginger_Power_240mA,Process,,Y,Y,,,,,,,
PowerTransfer_BatCharging_360mA,Fwdl,,Y,Y,g3,,,,,,
Transition_Ginger_Read_360mA,Process,,Y,Y,g4,,,,,,
Transition_Ginger_Power_360mA,Process,,Y,Y,,,,,,,
PowerTransfer_BatCharging_480mA,Fwdl,,Y,Y,g5,,,,,,
Transition_Ginger_Read_480mA,Process,,Y,Y,g6,,,,,,
Transition_Ginger_Power_480mA,Process,,Y,Y,,,,,,,
```

Start Parallel Thread

Join Thread Point



Terminology

SetChargerParams	Set and verify charger top-off parameters
Set_Bud_Voltage	Set and verify bud top-off voltage limits

THANK YOU

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